

Executive Summary

Law enforcement vehicles today carry numerous networked devices – routers, computers, cameras, sensors, and peripherals – many of which expose IP or radio identifiers. Common in-car routers (e.g. Cradlepoint or Sierra Wireless) often use default LAN IPs (Cradlepoint's primary interface is 192.168.0.1 ¹ ; Sierra's AirLink uses 192.168.13.31 ²) and broadcast SSIDs (e.g. "Cradlepoint_xxx" ³) or Bluetooth IDs for management. Rugged laptops/tablets (Panasonic Toughbooks, Getac, etc.) and in-car printers (Brother, Zebra, Printek) typically appear on the local network via DHCP or Wi-Fi, with identifiable hostnames or OUIs (e.g. Panasonic or Zebra MAC prefixes). Dedicated devices – dash cameras, ALPR/ANPR units, body-camera docks, and vehicle telematics modules – often have static IPs or link-local addresses and unique identifiers (serial numbers, MACs, RFID tags). Identification methods include scanning for IP ranges (e.g. Motorola 4RE DVRs use 192.168.97–99.x), examining MAC OUIs (Motorola Solutions OUI 00:18:85 ⁴ , Axon & WatchGuard MAC prefixes, etc.), mDNS/SSDP beacons, Bluetooth LE advertisements, QR/NFC labels, and VIN-linked asset registries. Procurement specs show typical models – e.g. Cradlepoint IBR1900 routers, Axon Fleet3 and WatchGuard dashcams, Vigilant or Idemia ALPR cameras ⁵ .

Practical discovery uses passive methods (Wi-Fi sniffing with Kismet/Airodump, Bluetooth scanning, RFID/NFC readers) and active probes (Nmap scans of common subnets, SNMP or HTTP fingerprinting, mDNS/UPnP enumeration). Tools like Nmap, Fing, Shodan (for internet-exposed units), and specialized radio scanners can help. However, legal constraints are stringent: one must generally obtain authorization before scanning police gear ⁶ . Unencrypted Wi-Fi beacons or SSIDs may be passively collected, but intercepting actual data can implicate wiretap laws ⁷ . Agencies' security manuals (e.g. FBI CJIS) require strict controls on wireless APs.

Security analyses reveal vulnerabilities in police-grade devices: e.g., Axon reported that older docking stations (firmware ≤ 3.1) allowed HTTP configuration changes without login ⁸ , and researcher disclosures detail unauthenticated admin access on Axon docks ⁹ . Security audits of body cameras found Wi-Fi radios broadcasting static SSIDs and MACs that encode make/model ¹⁰ , lacking update/file signing, enabling footage manipulation. Dashcams and ALPR systems have produced forensic artifacts (video, GPS, timestamp overlays) that require OCR/transcription ¹¹ .

We recommend lawful reconnaissance: use visual asset inspection (QR/NFC tags, VIN decoders), consult public procurement docs, and restrict to passive observation of openly broadcast data (SSID beacons, Bluetooth advert packets, RFID tags). Do not attempt to join or scan police networks without permission. The comparative table below summarizes each device category by identification cues, network behavior, discoverability, and forensic artifacts.

```
graph LR
  subgraph Patrol_Vehicle_Network [Patrol Vehicle Network]
    Router["In-Vehicle Router<br/>(e.g. Cradlepoint)"]
    MDT["Mobile Data Terminal (Laptop/Tablet)"]
    DashCam["Dash Camera (e.g. Axon Fleet)"]
  end
```

```

ALPR["ALPR Camera"]
Printer["In-Car Printer"]
Telematics["Telematics/GPS Module"]
BodyDock["Bodycam Docking Station"]
Evidence["Evidence Locker<br/>(RFID/QR)"]
Router --> MDT
Router --> DashCam
Router --> ALPR
Router --> Printer
Router --> Telematics
Router --> BodyDock
end
Internet -. LTE/4G .-> Router

```

1. Networking and Communications Devices

- **In-Car Routers/LTE Modems:** Rugged routers (Cradlepoint IBR series, Sierra Wireless AirLink MP70/LX) provide WAN backhaul and LAN/Wi-Fi. They typically use fixed LAN IPs (Cradlepoint's primary network is 192.168.0.1 ¹ ; AirLink uses 192.168.13.31 ²). These devices broadcast management SSIDs (e.g. "Cradlepoint_xy:zz" with MAC suffix ³) and may advertise Bluetooth BLE services for setup. MAC OUIs (assigned to Cradlepoint, Sierra Wireless, etc.) can be matched (e.g. Motorola Solutions OUI 00:18:85 covers many LTE devices ⁴). Routers often run DHCP on the LAN side, assigning IPs to connected devices. Procurement contracts show agencies installing specific models (e.g. "Cradlepoint modem (1900)" in patrol upfits ⁵). LTE modems embedded in routers or tethered smartphones usually have no local IP except their management interface; these appear only on the carrier network (sometimes identifiable via IMEI/VIN records in vehicle data logs).
- **Mobile Data Terminals (MDTs):** Rugged laptops/tablets (Panasonic Toughbook, Dell, etc.) connect via Ethernet or Wi-Fi to in-car networks. They often receive DHCP addresses from the router. Identification clues include hostnames (computer name broadcasts or mDNS), Windows Workgroup names, and Bluetooth OUIs (Panasonic Bluetooth radios). They may have asset QR/NFC tags or serial numbers linked to VIN in police inventory. No standard IP range – they follow the LAN's DHCP pool. Ports/services: typical endpoints (RDP/3389 if enabled, SSH, HTTP management tools). Discoverability: if LAN-bridged, a network scan (ping or Nmap) might find them. Otherwise, look for public device discovery protocols (mDNS, NetBIOS) or active Wi-Fi SSIDs from tethered devices.
- **Vehicle Wi-Fi Access Points:** Some routers provide officer/passenger Wi-Fi. SSIDs often reflect the agency or router ID; some agencies disable this. If enabled, these APs broadcast beacon frames (detectable by passive Wi-Fi monitors). They usually use WPA2 with a pre-shared key or enterprise mode; default settings vary. Ports: typical AP management on 80/443. Security best practice (CJIS) requires strong encryption ¹² . Hotspots may be set to "hidden" but probe requests can reveal their existence. Passive sniffing or Wireshark can log beacon SSIDs.
- **Telematics/GPS Modules:** Fleet telematics (e.g. Ford Pro, Verizon Connect, Mixtelematics) often plug into OBD-II or are OEM-installed. They communicate over cellular; some offer Bluetooth for configuration. No user-accessible IP; devices are identified via VIN, serial number, or mobile app pairing. Discoverable only by physically locating the module (no network scanning). Smart-city sensors (like gunshot detectors) might have known radio IDs but are outside vehicle networks.

2. Surveillance and Video Systems

- **Dash Cameras (In-Car Video):** Mobile DVRs record front/rear/interior. Examples include Axon Fleet, WatchGuard, and Panasonic 4RE systems. Many integrate with the vehicle router to stream video or upload evidence. Motorola's 4RE uses reserved subnets 192.168.97.x/98.x/99.x for its cameras. Default settings often place cameras on 169.254.x.x if DHCP absent. MAC OUIs reveal vendors (Axis Camera models used in some in-car systems, FLIR for infrared, etc.). Open ports may include RTSP (554), HTTP admin interfaces (80/443 or custom ports like 8080). mDNS/SSDP/UPnP broadcasts can list cameras ("AXIS...", "FLIRCam" names). Known vulnerability: Axis IP cameras used in Axon Interview stations had remote-control flaws ¹³ (suggesting similar IP cams in cars could be at risk). Forensics: dashcam data (video files, GPS/log overlay) generate artifacts – time/date stamps, speed, coordinates – requiring OCR and geolocation processing ¹¹. Investigators extract MPEG files and logs from SD cards or memory.
- **Automatic License Plate Readers (ALPR/ANPR):** Roof-mounted LPR cameras (e.g. Idemia/MorphoTrak, Genetec AutoVu, Vigilant) capture plates and send data to backend. They connect via Ethernet to the patrol network. Typically they use static LAN IPs (often manually set) or self-assigned link-local addresses. Some use proprietary software modules (browsable via HTTP/HTTPS, e.g. WebUI ports). MAC OUIs correspond to the camera vendor (e.g. Axis/Genetec). Discovering ALPR units requires scanning local subnets or sniffing packet flows (often encrypted over TLS). These systems rarely broadcast Wi-Fi or Bluetooth. They have high discoverability difficulty unless one can tap the LAN. Vulnerabilities are seldom public, but industry disclosures (e.g. exposure to malware in smart city LPR networks) underline that any IP camera can be compromised. The forensic footprint includes databases of plate hits (with timestamps/GPS) usually stored on central servers.
- **Body-Worn Camera Docking Stations:** Docks (e.g. Axon/Taser Dock) upload video from bodycams to Evidence.com. Early versions had built-in web servers on the LAN (accessible via default IP, often 192.168.x.x ⁸) and default credentials. Researchers found an auth bypass: unauthenticated HTTP POSTs to "/lua" endpoints could reconfigure docks, reset them, or de-register them ⁹. Axon's own advisory confirmed that older Dock firmware (≤ 3.1) allowed config changes without login ⁸. Newer Axon Body 3 docks behave as unmanaged switches: **they have no LAN IP** ¹⁴ – each bay simply passes Ethernet to the camera. Thus, the dock itself is invisible on the network except via the camera. Each dock bay does have a MAC printed on its end ¹⁴, used for DHCP reservations. Active scanning: one could attempt to reach old docks on 80/443. After 2016, Axon pushed a 3.2 firmware to fix the flaw ⁸. Forensics: docks keep logs of camera serials and upload timestamps in the Evidence.com portal; the dock hardware leaves little artifact beyond ethernet switch ARP table entries.
- **Vehicle Intercom/Siren Controllers:** Not usually networked via IP, so typically out of scope. Some new systems (mobile mesh radio, Bluetooth PAC devices) exist but are minor.

3. Accessory and IoT Devices

- **Mobile Printers:** Thermal printers (Brother PocketJet, Zebra ZQ series, Printek) may connect by USB, Bluetooth, or Wi-Fi. Bluetooth printers advertise a name (often including model). Wi-Fi printers create an AP (e.g. default SSID "Printer_XXXX") or join the LAN. Default network-mode IPs vary by model (some like Zebra can use 192.168.x.x AP mode). OUIs for Brother or Zebra appear on their MACs. Common ports: 9100 (LPD/JetDirect), 80/443 for management, Bluetooth SPP/RFCOMM. Discoverability: Bluetooth inquiries might reveal them; Wi-Fi can be found via probing (if open) or by

listening for SSIDs. Forensics: print logs (if any) are usually internal; the main artifact is the printed ticket.

- **Evidence Lockers/Key Safes:** Some patrol cars carry small smart lockers (weapon safes, evidence bins) from vendors (e.g. ProGuard, Viper, AssetTracer). These may use RFID or QR-encoded tags for item tracking. The locker itself is not typically networked (it's a mechanical lock). However, departments increasingly tag assets with RFID or barcode ¹⁵. For example, RFID systems track weapons, vehicles, and evidence ¹⁵. Identification method: scan the tag with a handheld reader or phone app to read serial/ID. No IP or MAC exposure. Forensics: RFID system logs chain-of-custody, but the locker leaves no digital trace except possibly an internal event log (not accessible externally).
- **Vehicle Sensors (Miscellaneous IoT):** Some modern patrol cars have OBD-II telematic dongles or body sensors (accelerometers, cameras for collision avoidance). These usually talk to the vehicle's bus, not to an IP network. The telematic dongles (for fleet telematics) may use Bluetooth for setup, but otherwise report via cellular. Tire pressure, seatbelt sensors, etc., are internal CAN bus and not externally visible. Gunshot detectors or environmental sensors are not generally mounted in cars. Therefore, most "IoT" in a cruiser is just the vehicle's own telemetry.

4. Identification and Network Fingerprinting Methods

For each device category, key discovery cues include:

- **IP Address and Subnet:** Many devices use fixed subnets. For instance, Motorola's 4RE video system uses 192.168.97-99.x. Cradlepoint routers use 192.168.0.1 ¹. Bodycam docks (older) might be on 10.x. Knowing typical defaults lets one target specific ranges with ping/nmap. Devices set to DHCP may grab 192.168.x.x, 172.16.x.x, or 10.x.x.x from the router's pool. Scanning these common ranges is a first step.
- **MAC/OUI:** Examining MAC addresses (from ARP or passive capture) reveals vendor. For example, Motorola Solutions owns OUI 00:18:85 ⁴, Cradlepoint's OUIs (e.g. 28:94:2A) can be looked up, as can Sierra Wireless (e.g. 00:0B:99). Bluetooth also has company IDs. Identifying OUIs can narrow device type (e.g. any 00:18:85 MAC in-car is likely Motorola/WatchGuard video gear). Many IP cameras or IoT sensors use known OUIs.
- **Ports and Services:** Common open ports hint at device type. Routers often have web (80/443/8080), SSH (22), SNMP (161), or MQTT. LPR cameras may run HTTP(S) for plate data. Dashcams may offer FTP/SSH for logs. Printers usually listen on 9100, 515, 80 (web config). UPnP/SSDP discovery (listening on 1900 multicast) can pick up any "simple device" broadcasts (e.g. networked cameras, media servers). mDNS/Bonjour (224.0.0.251) sometimes reveals hostnames like "AxonFleet" or "PanasonicEthernet" if enabled.
- **SSIDs / Wi-Fi Beacons:** Wireless equipment often uses identifiable names. Cradlepoint routers' SSIDs include the last 3 bytes of MAC ³. Bodycams or printers running hotspots may use default SSID formats. Passive Wi-Fi scanning (beacons/probes) can map these SSIDs to MACs and thus to device types. For example, a surveillance camera AP might be "4RE camera_01".
- **Bluetooth LE Advertising:** Many devices (even if not primarily Bluetooth) use BLE for maintenance. Modern bodycams (Axon Body3) don't, but older cameras or printers might. BLE advertisements often include a device name (e.g. "SHN-CAM123"). Tools like `bleah`, `BTScanner` or smartphone apps can list these. Note: Bluetooth scanning is limited by range (<50m) and must typically be active (inquiry).
- **QR/NFC Tags and Labels:** Manufacturers often attach QR codes or NFC tags with serial/model (for inventory). Evidence or docking stations might have QR codes linking to asset records. While not

network-scanning, a field researcher could photograph or scan these tags to identify device make/model and cross-reference manuals (some user guides list default IPs or Wi-Fi credentials).

- **VIN/Serial Associations:** Police tech manuals often associate equipment with vehicle VIN. Forensic acquisition may include scanning the vehicle VIN barcode (found under the dash) and then querying fleet databases to infer installed systems. Reverse-engineering VIN lookup APIs (e.g. NHTSA) can reveal OEM telematics options.
- **mDNS/UPnP Signatures:** Use tools like `dns-sd` or UPnP clients to browse local network services. Devices may advertise `_http._tcp.local` with a friendly name (e.g. “Axon Dock”) or SSDP UPnP info. Passive listening on 239.255.255.250 (SSDP) can catch Universal Plug and Play announcements, which some IP devices use for discovery.

5. Scanning and Discovery Techniques (with Constraints)

Passive Techniques: Without transmitting, one can still glean info.

- *Wi-Fi sniffing:* A laptop or smartphone with wireless in monitor mode can capture beacon frames and probe responses from in-car APs or cameras. Tools: Kismet, Airodump-ng, or Wireshark. Look for SSIDs like “Cradlepoint_xx”, “Fleet-AP”, etc., and note BSSIDs (MACs). Also capture any broadcast mDNS/SSDP packets.
- *Bluetooth LE Scanning:* Use BLE scanners (e.g. Nordic nRF Connect app, `hcitool lescan`) to detect BLE advertising devices. Although many police devices may not broadcast BLE, some may use it for pairing. Recording BLE advertisements over time can reveal device presence and identifiers.
- *RFID/Barcode Reading:* For evidence lockers and inventory, passive RFID readers or smartphone cameras can scan tags. NFC tags (if present) can be read with a phone app. This requires physical proximity inside the vehicle.
- *Visual Inspection:* Photograph device labels (make/model, FCC ID) and serial numbers. Many manufacturers list default credentials or IPs in manuals (e.g. “See label for login”³). Check asset tags linking VINs to equipment.

Active Techniques (with Authorization): If permitted, more direct probing is possible.

- *Network Scanning:* Connect to a benign interface (e.g. the router’s guest Wi-Fi if open, or tether to router) and run Nmap or Fing over likely subnets (192.168.0.0/24, 172.16.0.0/16, etc.). Use ping sweeps and port scans to identify hosts. For example, scanning 192.168.97.1–254 might reveal Motorola camera IPs. Use `-O` for OS fingerprinting to distinguish routers vs PCs. Run `nmap -sV` to identify services (e.g. “Axis Audio Control”).
- *SNMP/UPnP:* Some routers or printers may have SNMP enabled (default community “public”). An `snmpwalk` could list connected device MACs and firmware. UPnP SSDP scans can find DLNA devices or IP cameras.
- *mDNS/Bonjour:* On the LAN, browse for Bonjour services (e.g. `dns-sd -B _http._tcp`, `_ipp._tcp` for printers). Many IoT devices implement Bonjour to make setup easy.
- *Telnet/SSH/HTTP:* Try logging into devices using default credentials (e.g. Cisco ASA guides warn default admin on Axon Dock⁸; older routers often “admin/admin”). Use tools like Hydra for brute-forcing if authorized.
- *Wi-Fi Association:* If an AP is open or Wi-Fi Direct (bodycam hotspot), one could join it and enumerate its clients or services. This is often against policy.

Constraints & Best Practices: Unauthorized scanning is legally risky. Nmap's author advises always obtaining written permission ⁶. In the U.S., the CFAA and wiretap laws could be construed to cover unauthorized network access or interception. Notably, a court ruled that *passively listening* to unencrypted Wi-Fi (beacons) is “readily accessible” and thus not a wiretap violation ⁷ – but this is contested. At minimum, scanning police IT without consent may violate internal policies or state laws. We therefore recommend *only passive observation of visible identifiers* (SSIDs, BLE, RFID, VIN) and *no payload capture or injection* without explicit authorization. Document devices by sight (label/QR) rather than probing when in doubt. Follow NIST SP800-115 guidance on minimizing intrusion: start with passive, limit active probes, log all actions for audit.

6. Known Vulnerabilities and Forensic Findings

- **Body-Worn Camera Security:** Researchers at DefCon (2018) analyzed five police bodycams and found severe flaws ¹⁰. Four models broadcast static Wi-Fi SSIDs/MACs containing make/model, enabling long-range tracking of officers ¹⁰. The cameras lacked firmware-signing and strong crypto, so attackers (within Wi-Fi range) could download, alter, or delete footage and re-upload it undetected. E.g. a Wi-Fi attacker could log in as “admin” on affected models, view/modify media and file structures ¹⁰. (Axon-brand cameras were *not* tested in that study, but similar principles apply.)
- **Docking Station Flaws:** In 2016 Axon disclosed (and researchers confirmed) that Axon Dock v3.1 had an HTTP interface flaw ⁸ ⁹. By sending crafted requests (e.g. to `/lua/set-passwd.lua`), an attacker on the LAN could change the admin password, reconfigure networks, deregister the dock from Evidence.com, or initiate a reboot ⁹. This allowed unauthorized control of the dock (and potential firmware upload to it or attached cameras). Axon remedied this in v3.2, but agencies with unpatched docks were vulnerable. No known external incidents were reported, but this illustrates that “non-IP” equipment can expose HTTP endpoints.
- **Dashcam/ALPR Forensics:** Academic work shows dashcams leave rich forensic traces: embedded timestamps, GPS, speed, and even license plate images ¹¹. A 2023 study proposed extracting overlay text via OCR, and audio cues via speech-to-text, to reconstruct events ¹¹. Crash-warning logs (lane-departure alerts) are also recorded. Mobile ALPR units similarly log plate images and hit data, which investigators often recover from central servers. While not “vulnerabilities,” these artifacts mean investigators should carefully image SD cards and log files. Specialized tools (e.g. Magnet AXIOM, Belkasoft) are only now adding dashcam modules, per that study’s findings.
- **Other Reports:** Aside from docks and cameras, some agencies have found misconfigured Wi-Fi on older equipment (e.g. default SSIDs/passwords on fleet modems), but few public disclosures exist. Shodan and Cert advisories can reveal exposed police networks (e.g. open router admin pages), though CJIS rules mandate firewalls. Wireless dispatch consoles (e.g. Motorola APX) are typically closed systems with no public radios, so they seldom appear. However, any open access point (even a misguided public Wi-Fi in station cars) is a risk – hence policy often requires disabling unnecessary radios ¹² ⁶.

7. Recommendations and Best Practices

- **Passive Identification:** Emphasize non-intrusive methods. Use binoculars or a camera to read device labels, SSIDs printed on stickers, and QR/NFC asset tags. Maintain a catalog of known manufacturers and models (e.g. a “fingerprint” list of OUIs or SSID prefixes for Cradlepoint, Axon, WatchGuard, etc.). Where safe/allowed, observe radio signals (SSIDs, BLE) without joining networks.

- **Documentation:** Keep a log of vehicle VINs and optional equipment. Public procurement databases (e.g. SAM.gov, local bid documents) often list installed kits (as we saw ⁵). Use this to anticipate device types. Take photos of interior consoles and compartments; these often contain routers and docking stations.
- **Authorized Scans:** If part of an audit with permission, conduct scheduled scans during maintenance windows. Isolate the vehicle's network (pull fuse or use a shielded testbench) to avoid interfering with live services. Use VLANs or network taps. Only scan common ports, and immediately cease any intrusive action (e.g. logout, rename admin accounts) on finding default credentials. Record all findings for chain-of-custody (timestamps, tool outputs).
- **Legal Compliance:** Follow NIST/FBI guidelines for wireless security. Treat any unencrypted channel as non-private (best practice: use spectrum analyzers to detect unauthorized APs). Do not deploy deauth or jamming tools near law enforcement. If any doubt, consult legal counsel: for instance, benign scanning of broadcast SSIDs is generally permissible ⁷, but connecting to a secure police AP without authorization could breach the Computer Fraud and Abuse Act ⁶.
- **Forensics Handling:** When collecting data (e.g. pulling an SD card), use standard DFIR techniques. For devices like dashcams, image storage media read-only. Document device firmware versions (noting known CVEs) and network logs. For bodycams, ensure evidence chain is maintained even if vulnerabilities are found (update firmware and note any tamper-evidence). Collaborate with manufacturers when possible (Axon provided firmware updates for the Dock issue ⁸).

8. Device Comparison Table

Device Type	Identification Methods	Network Exposure / IP	Typical IP Behavior	Discoverability	Forensic Artifacts
In-Car Router (Cradlepoint, Sierra, etc.)	Default LAN IP 192.168.0.1 or 192.168.13.31 ¹ ² ; SSID (includes MAC bytes) ³ ; MAC OUI from vendor; model on sticker.	Always on 4G; provides LAN/Wi-Fi; static LAN IP by DHCP.	Mostly static (or reserved via DHCP); often gateway at .1.	Moderate: AP SSID visible; MAC OUIs known.	Router logs (connected clients); SIM/account data.
MDT/ Computer (Toughbook, iPad, etc.)	Hostname via NetBIOS/mDNS; Bluetooth OUI (e.g. Panasonic BT); device model via management interface; asset tag.	Joined to vehicle LAN (Ethernet or Wi-Fi); DHCP.	Dynamic from router; often Windows hostnames (e.g. "OFFICER-PC").	Moderate: LAN scan detects IP; Bluetooth may advertise.	OS logs, Wi-Fi network configs, local files.

Device Type	Identification Methods	Network Exposure / IP	Typical IP Behavior	Discoverability	Forensic Artifacts
Dash Camera (FleetCam)	MAC OUI (Axis, Panasonic, Digital Ally); mDNS name; static DHCP; camera serial on label.	Connected via router Ethernet; sometimes 4G; often static IP (e.g. 192.168.97.x).	Often static or specific subnet (Motorola reserves 97-99.x); uses RTSP/HTTPS.	Hard: no Wi-Fi unless AP mode; LAN scan needed.	Video files, metadata (timestamp, GPS), log files.
ALPR Unit	Camera OUI (FLIR, Axis); no SSID; may broadcast HTTP/RTSP; IP from DHCP or static (commonly 169.254.x if isolated).	Wired to router; cellular uplink. No open Wi-Fi.	Usually static/dhcp on car LAN; uplinks plate data via LTE.	Very hard: hidden behind router; only passive LAN sniff or licensed frequency scanning.	Plate capture logs, GPS/ time stamps in ALPR server.
Bodycam Docking Station	(Old) web interface on HTTP; default IP (e.g. 192.168.x). Axon Dock: firmware label, fixed MAC per bay ¹⁴ ; no SSID.	Ethernet connection; DHCP via switch; new docks have <i>no IP</i> ¹⁴ .	Each bay has separate MAC (visible as DHCP client); no static IP.	(Old) possible if web open; (New) dock invisible – only bay MACs.	Upload logs to Evidence.com; internal certificates (Axon Body3).
Printers (mobile)	Bluetooth name (e.g. “ZebraPrinter”); Wi-Fi SSID (“Printer_xxx”); MAC OUI (Brother, Zebra).	USB/ Bluetooth to MDT; Wi-Fi AP or client.	If Wi-Fi: usually self-assigned (192.168.x.x) in AP mode.	Easy: Wi-Fi SSID; Bluetooth scan picks up printer name.	Print history (limited); if wifi, web logs.
Wi-Fi Access Point (hotspot)	SSID name (often agency-branded); BSSID MAC; Bluetooth (for some routers).	Wi-Fi radio on router; WPA2 encryption.	DHCP or NAT by router.	High: SSID beacon visible; hidden SSID still leaks probe responses.	Client association logs in router.

Device Type	Identification Methods	Network Exposure / IP	Typical IP Behavior	Discoverability	Forensic Artifacts
Vehicle Telematics/ GPS	VIN, device serial; Bluetooth for config (if any); OBD-II code.	Cellular-only (no LAN IP); some OBD units may create local Wi-Fi.	Interfaces with car systems; IP only via back-end.	Very hard: usually no beacon; only detect if running a Bluetooth setup mode.	GPS logs, speed, engine data on server.
Evidence/ Weapon Locker	RFID tag or QR label scanned by reader; lock serial.	None (mechanical or RFID only).	N/A.	Low: must physically scan tag.	Scan history (if logged in RFID system); no network footprint.
IoT Sensors (misc.)	Sensor ID (like sensor serial), OUI if networked (rare).	Usually Bluetooth or proprietary (LPWAN); few IP interfaces.	If Wi-Fi, likely fixed config (e.g. dash light cameras).	Low: BLE scan if present; Wi-Fi sensors broadcast.	Sensor logs (internal), event recordings.

Sources: Manufacturer manuals and security advisories provide defaults (e.g. Cradlepoint 192.168.0.1 ¹, Axon Dock vulnerabilities ⁸ ⁹). Forensic studies (e.g. dashcam OCR ¹¹) and industry reports (Wired bodycam analysis ¹⁰) inform discoverability and artifact expectations. The above table and descriptions synthesize these findings into a unified view of typical police-vehicle device deployments. All scanning/discovery should follow legal guidance ⁶ and agency policy.

¹ ³ [cradlepoint_cor_ibr11x0_manual_05.18.15](https://blog.magnumelectronics.com/wp-content/uploads/2015/06/cradlepoint_ibr11x0_manual_05-18-15.pdf)

https://blog.magnumelectronics.com/wp-content/uploads/2015/06/cradlepoint_ibr11x0_manual_05-18-15.pdf

² [AirLink MP70 Sierra Wireless Default Router Login and Password](https://modem.tools/router/sierra-wireless/airlink-mp70)

<https://modem.tools/router/sierra-wireless/airlink-mp70>

⁴ [MAC address 00:18:85 | MAC Address Lookup](https://maclookup.app/macaddress/001885)

<https://maclookup.app/macaddress/001885>

⁵ [pagosasprings.co.gov](https://www.pagosasprings.co.gov)

<https://www.pagosasprings.co.gov/DocumentCenter/View/367/ITB-2026-PW-01-New-Police-F-150-Responders>

⁶ [Legal Issues | Nmap Network Scanning](https://nmap.org/book/legal-issues.html)

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⁷ [District Court Rules that the Wiretap Act Does Not Prohibit Intercepting Unencrypted Wireless Communications - The Volokh Conspiracy](https://volokh.com/2012/09/06/district-court-rules-that-the-wiretap-act-does-not-prohibit-intercepting-unencrypted-wireless-communications/)

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- 8 **AXON DOCK AUTHENTICATION BYPASS - SECURITY NOTIFICATION AXON-1601 - Axon.com**
<https://www.axon.com/security/advisory-axon-1601>
- 9 **Bugtraq: Taser Axon Dock (Body-Worn Camera Docking Station) v3.1 - Authentication Bypass**
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- 11 **Digital Forensic Analysis of Vehicular Video Sensors: Dashcams as a Case - PMC**
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- 14 **Axon Body 3 Dock Installation Guide**
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- 15 **RFID for Asset & Equipment Tracking in Police and Fire Departments**
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